

Integrated Library Management System (CDM ILMS)

Standard System Diagrams (ERD, DFD Level 0, DFD Level 1, Flowcharts) & Technical Defense Guide

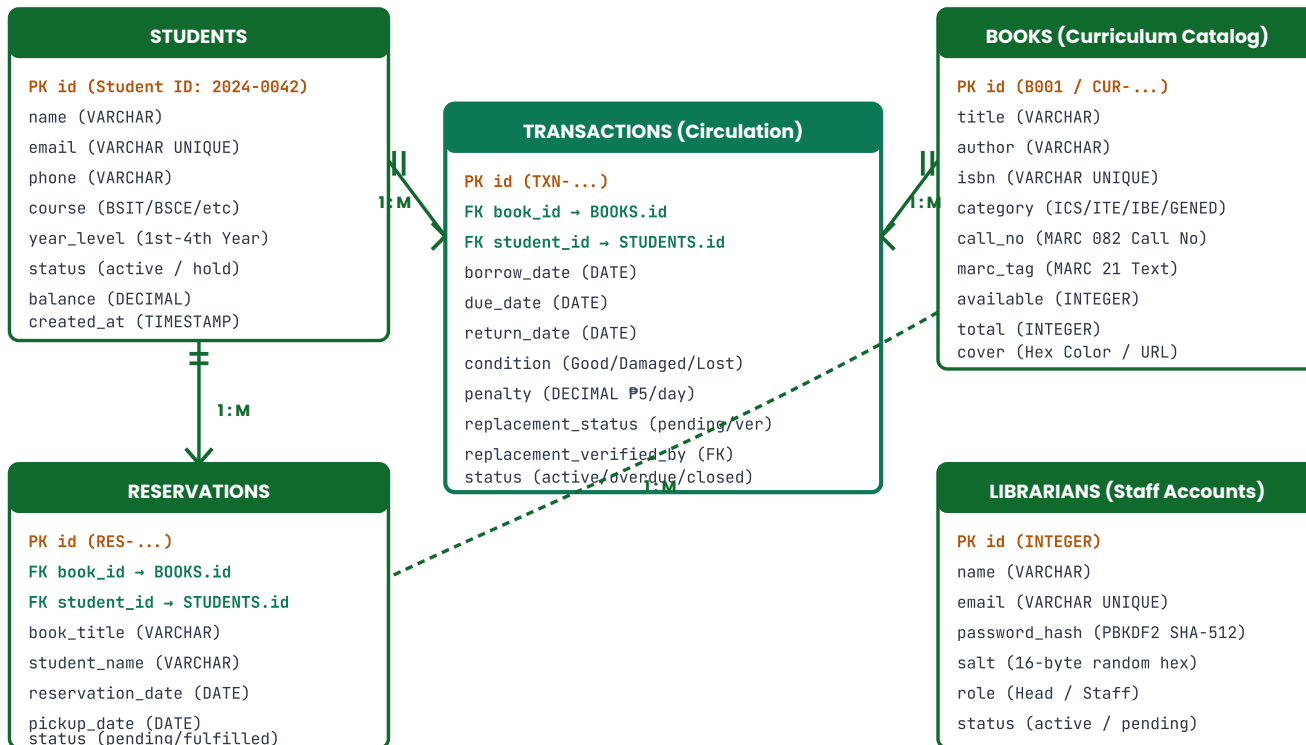
Architecture: Tri-Platform (Desktop, Mobile & Web OPAC) **Database:** SQLite 3 (library.db) **Standard Notation:** Chen / Crow's Foot ERD, Gane-Sarson DFD, ANSI Flowcharts

1. ENTITY RELATIONSHIP DIAGRAM (ERD) — STANDARD RELATIONAL SCHEMA

The CDM ILMS database consists of **6 normalized tables** in SQLite (library.db). Entities use standard primary keys (PK), foreign keys (FK), and Crow's Foot cardinality notations (1:Many):

Relational ERD with Crow's Foot Cardinalities

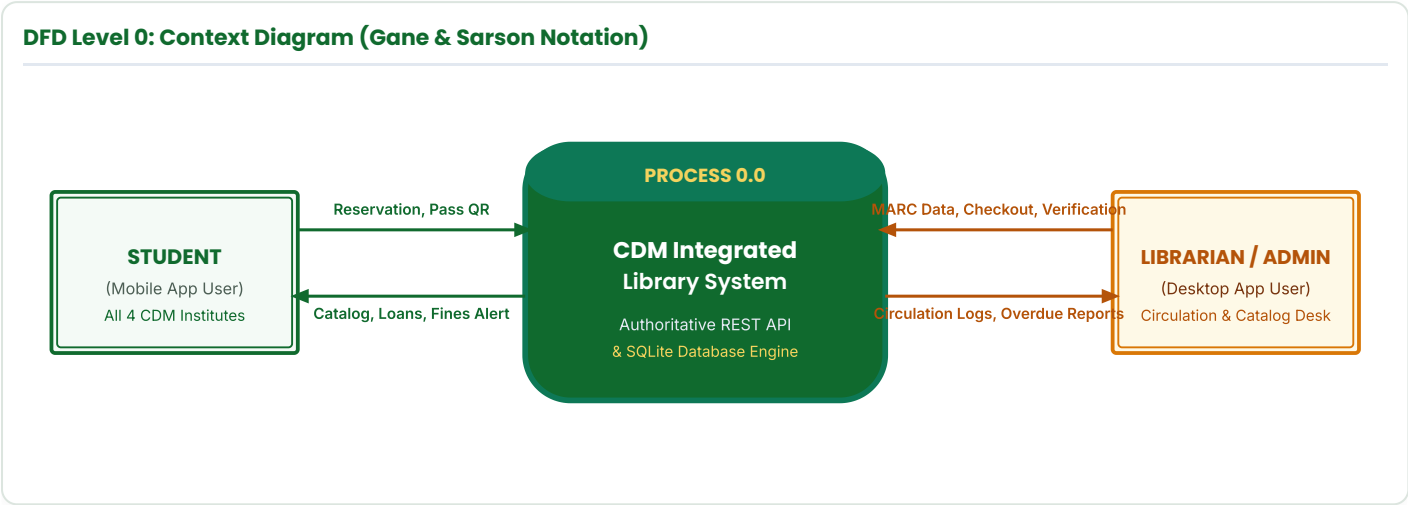
■ Core Entity ■ Transaction Entity PK Primary Key FK Foreign Key



2. DATA FLOW DIAGRAMS (DFD) – GANE & SARSON STANDARD NOTATION

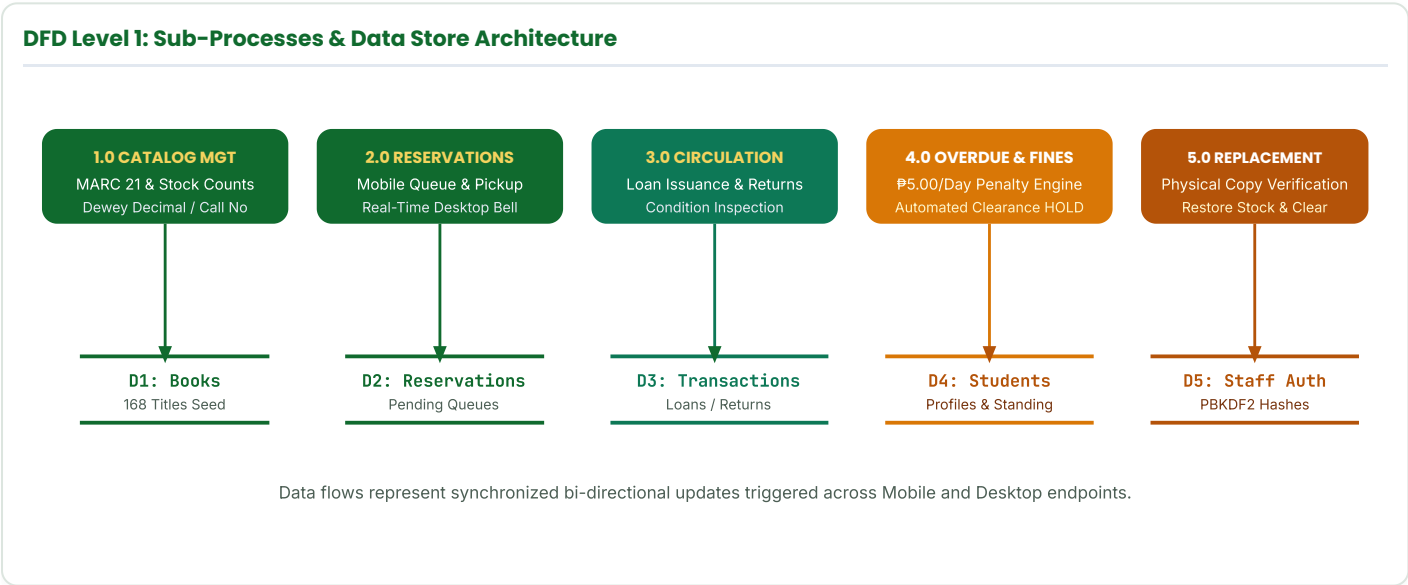
2.1 DFD Level 0: Context Diagram

The Level 0 Context Diagram establishes the complete system boundary, showing information flows between external entities and the centralized CDM ILMS core process:



2.2 DFD Level 1: Sub-Process Decomposition Diagram

The Level 1 DFD decomposes the system into 5 primary sub-processes and 5 standard open-ended Data Stores:

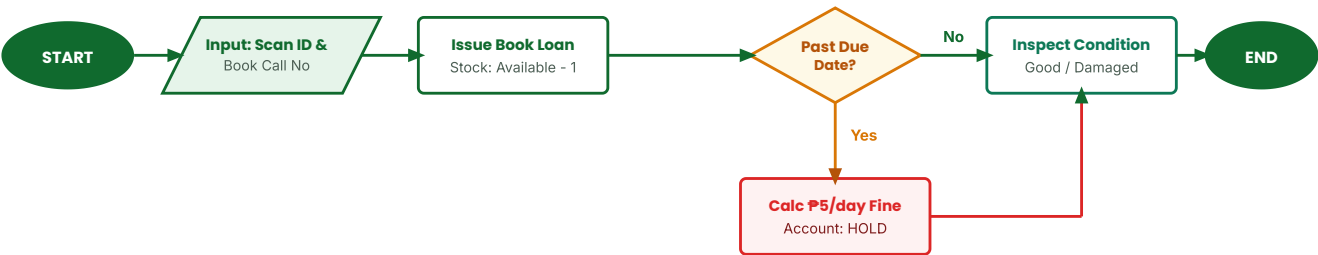


3. SYSTEM FLOWCHARTS — STANDARD ANSI/ISO SHAPES

Standard flowchart shapes: Terminators (Ovals), Processes (Rectangles), Decisions (Diamonds), and Input/Output (Parallelograms).

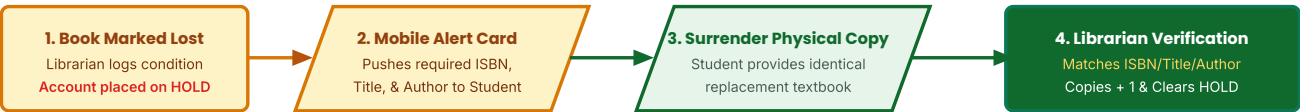
3.1 Main Circulation & Loan Lifecycle Flowchart

Circulation Desk & Penalty Enforcement Flowchart



3.2 Mandatory Physical Lost Book Replacement Workflow (CDM Library Policy)

Lost Book Physical Replacement & Verification Workflow



4. ORAL DEFENSE PRESENTATION SCRIPT & KEY PANEL Q&A

-  **5-Step Live Demonstration Script for the Defense Panel:**
- 1. **Tri-Platform View:** Open Desktop (:5173), Mobile (:5175), and Web OPAC (:5002/promo). Explain that all 3 frontends communicate with one authoritative Node.js API and SQLite database.
 - 2. **168-Book Academic Catalog:** Filter catalog by Institute (ICS) and 4th Year. Click 'MARC Record' to showcase international MARC 020 (ISBN), 082 (Dewey Call No), and 245 (Title) tags.
 - 3. **Real-Time Mobile Reservation:** On Mobile, select a book, choose a target pickup date, and tap 'Reserve Book'. Show Desktop notification bell badge incrementing immediately.
 - 4. **Loan Issuance & Overdue Engine:** Issue loan to 2024-0042. Show Mobile 'My Library' tracking live countdowns and calculating daily ₱5.00 penalties.
 - 5. **Physical Replacement Workflow:** Mark loan as 'Lost' on Desktop $\rightarrow$ Show Mobile amber alert with ISBN $\rightarrow$ Click 'Verify Physical Replacement' on Desktop $\rightarrow$ Show account restored to 'Cleared'!

Top Defense Panel Questions & Model Answers

- Q1: Why did you choose SQLite over MySQL or MongoDB?**
Answer: "We selected SQLite 3 (via better-sqlite3) because it provides zero-latency synchronous execution, requires zero server administration, and encapsulates the entire relational schema inside a single portable file (library.db). For a college campus intranet, SQLite handles over 50,000 queries per second with minimal memory footprint."
- Q2: How does the system handle real-time synchronization between Mobile and Desktop?**
Answer: "Synchronization operates through our centralized Express REST API endpoints (/api/books, /api/student-portal/:studentId, and /api/reservations). When an action is taken on either client, the database state updates immediately and the clients refetch on action triggers."
- Q3: How do you prevent negative stock or race conditions during simultaneous checkouts?**
Answer: "We utilize SQLite atomic database transactions (db.transaction()). Deducting available inventory and logging the loan transaction execute as a single atomic unit. If any step fails, the entire transaction rolls back completely."
- Q4: Why enforce physical replacement instead of paying cash fines for lost books?**
Answer: "Under Colegio de Montalban library policy, specialized curriculum textbooks are difficult to repurchase quickly with cash. Requiring students to supply the exact physical copy (matching ISBN, Title, and Author) guarantees that institutional inventory and academic reference collections remain intact for succeeding student batches."
- Q5: How do you secure staff passwords and prevent brute-force attacks?**
Answer: "We hash passwords using PBKDF2 SHA-512 with a unique 16-byte random salt per user (10,000 iterations) and validate using constant-time comparison (crypto.timingSafeEqual). Login endpoints are protected by an in-memory sliding-window rate limiter (authLimiter) capping requests at 15 attempts per 15 minutes."

-  **Defense Demonstration Credentials & Ports:**
- **Librarian Desktop:** http://localhost:5173 | **Login:** admin@cdm.edu.ph / admin123
 - **Student Mobile App:** http://localhost:5175 | **Sample Student ID:** 2024-0042 (Jay Deguzman, BSIT 4th Year)
 - **Public Web Portal:** http://localhost:5002/promo | **Central API Status:** http://localhost:5002/api/books